COMPLETION IS NEITHER LEFT NOR RIGHT EXACT IN GENERAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The statement's slogan environment is kept verbatim. The second paragraph's load-bearing ingredient (noncompleteness of R/J) is the delivered card L48305 in the same unit pool.

Editorial note. The label lemma-completion-not-exact names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Context span 746-771, first paragraph: R = k[t], I = (t), P = K = direct sum over N of R, M = direct sum of R/(t^n); the short exact sequence 0 -> K -> P -> M -> 0 with first map multiplication by t^n on the nth summand; after completion, xi = (t^2, t^3, t^4, ...) in P^wedge maps to zero in M^wedge but is not in the image of K^wedge -> P^wedge since that would be the image of (t, t, t, ...) which is not an element of K^wedge
- Context, second paragraph: in the situation of Lemma noncomplete-quotient (delivered card L48305) the short exact sequence 0 -> J -> R -> R/J -> 0 does not remain exact after I-adic completion: with f a generator of J the map R -> J^wedge is surjective with image of J^wedge -> R equal to (f) = J, but R/J noncomplete means the kernel of the surjection R -> (R/J)^wedge is strictly bigger than J (Algebra, completion-generalities and quotient-complete); by the same token R -> R -> R/(f) -> 0 does not remain exact on completion

Source reference. Original lemma 0315: sources/stacks-chow/tags.txt, source lines 3972–3972. Prerequisite pointer invoked at 808; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 031A: sources/stacks-chow/tags.txt, source lines 3977–3977. Prerequisite pointer invoked at 809; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 05JE: sources/stacks-examples/pinned-original-source.tex, source lines 730–730. Prerequisite pointer invoked at 800; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:examples.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

A quick example is the following. Suppose that $R = k[t]$. Denote $M^\wedge$ the completion of an R-module with respect to $I = (t)$. Let $P = K = \bigoplus_{n \in \mathbf{N}} R$ and $M = \bigoplus_{n \in \mathbf{N}} R/(t^n)$. Then there is a short exact sequence $0 \to K \to P \to M \to 0$ where the first map is given by multiplication by t^n on the nth summand. We claim that $0 \to K^\wedge \to P^\wedge \to M^\wedge \to 0$ is not exact in the middle. Namely, $\xi = (t^2, t^3, t^4, \ldots) \in P^\wedge$ maps to zero in $M^\wedge$ but is not in the image of $K^\wedge \to P^\wedge$, because it would be the image of $(t, t, t, \ldots)$ which is not an element of $K^\wedge$.

A "smaller" example is the following. In the situation of Lemma 05JE the short exact sequence $0 \to J \to R \to R/J \to 0$ does not remain exact after completion with respect to I. Namely, if $f \in J$ is a generator, then $f : R \to J$ is surjective, hence $R \to J^\wedge$ is surjective, hence the image of $J^\wedge \to R$ is $(f) = J$ but the fact that R/J is noncomplete means that the kernel of the surjection $R \to (R/J)^\wedge$ is strictly bigger than J, see Algebra, Lemmas 0315 and 031A. By the same token the sequence $R \to R \to R/(f) \to 0$ does not remain exact on completion.

Lemma 0.1. *Completion is not an exact functor in general; it is not even right exact in general. This holds even when I is finitely generated on the category of finitely presented modules.*

Proof. See discussion above. □